

PRACTICAL COURSE WORKBOOK

Algorithmic Bias & Fairness

Evaluate technology decisions with evidence and accountability

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a documented risk and impact assessment
SKILLS	Bias, Fairness, Audit, ML Ethics, Disparate Impact, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Bias appropriately in a realistic, bounded task.
- Use Fairness appropriately in a realistic, bounded task.
- Use Audit appropriately in a realistic, bounded task.
- Use ML Ethics appropriately in a realistic, bounded task.

WHO THIS IS FOR

Product, policy and technical learners evaluating responsible AI.

BEFORE YOU BEGIN

- No specialist background required

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Algorithmic Bias & Fairness is a structured, practical course covering Bias, Fairness, Audit, ML Ethics, Disparate Impact. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Bias: Ethical foundations	1. Stakeholders with Bias 2. Harms with Fairness 3. Rights with Audit
02 Fairness: Evidence	1. Bias with ML Ethics 2. Measurement with Disparate Impact 3. Impact with Bias
03 Audit: Governance	1. Accountability with Fairness 2. Documentation with Audit 3. Oversight with ML Ethics
04 ML Ethics: Practice	1. Risk review with Disparate Impact 2. Participation with Bias 3. Monitoring with Fairness

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Bias: Ethical foundations

Apply Bias through stakeholders, harms, rights.

LESSON 01 / 18 MIN

Stakeholders with Bias

Learning objectives

- Explain stakeholders in the context of Algorithmic Bias & Fairness.
- Apply Bias to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

stakeholders, Bias, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a stakeholders result produced with Bias?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Harms with Fairness

Learning objectives

- Explain harms in the context of Algorithmic Bias & Fairness.
- Apply Fairness to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

harms, Fairness, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a harms result produced with Fairness?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Rights with Audit

Learning objectives

- Explain rights in the context of Algorithmic Bias & Fairness.
- Apply Audit to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

rights, Audit, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a rights result produced with Audit?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Fairness: Evidence

Apply Fairness through bias, measurement, impact.

LESSON 04 / 30 MIN

Bias with ML Ethics

Learning objectives

- Explain bias in the context of Algorithmic Bias & Fairness.
- Apply ML Ethics to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

bias, ML Ethics, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a bias result produced with ML Ethics?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Measurement with Disparate Impact

Learning objectives

- Explain measurement in the context of Algorithmic Bias & Fairness.
- Apply Disparate Impact to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

measurement, Disparate Impact, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a measurement result produced with Disparate Impact?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Impact with Bias

Learning objectives

- Explain impact in the context of Algorithmic Bias & Fairness.
- Apply Bias to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

impact, Bias, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a impact result produced with Bias?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Audit: Governance

Apply Audit through accountability, documentation, oversight.

LESSON 07 / 26 MIN

Accountability with Fairness

Learning objectives

- Explain accountability in the context of Algorithmic Bias & Fairness.
- Apply Fairness to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

accountability, Fairness, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a accountability result produced with Fairness?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Documentation with Audit

Learning objectives

- Explain documentation in the context of Algorithmic Bias & Fairness.
- Apply Audit to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

documentation, Audit, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a documentation result produced with Audit?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Oversight with ML Ethics

Learning objectives

- Explain oversight in the context of Algorithmic Bias & Fairness.
- Apply ML Ethics to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

oversight, ML Ethics, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a oversight result produced with ML Ethics?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

ML Ethics: Practice

Apply ML Ethics through risk review, participation, monitoring.

LESSON 10 / 22 MIN

Risk review with Disparate Impact

Learning objectives

- Explain risk review in the context of Algorithmic Bias & Fairness.
- Apply Disparate Impact to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

risk review, Disparate Impact, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a risk review result produced with Disparate Impact?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Participation with Bias

Learning objectives

- Explain participation in the context of Algorithmic Bias & Fairness.
- Apply Bias to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

participation, Bias, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a participation result produced with Bias?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Monitoring with Fairness

Learning objectives

- Explain monitoring in the context of Algorithmic Bias & Fairness.
- Apply Fairness to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, Fairness, ethics

Retrieval prompt

In Algorithmic Bias & Fairness, which evidence best supports a monitoring result produced with Fairness?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Algorithmic Bias & Fairness project using Bias, Fairness, Audit.

DELIVERABLE	A working Algorithmic Bias & Fairness artefact plus an evidence-based self-review.
TOOLS	A suitable Bias environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Bias.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Recommendation on the Ethics of AI

UNESCO - reviewed 2026-08-21

<https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>

AI Principles

OECD - reviewed 2026-08-21

<https://www.oecd.org/en/topics/ai-principles.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Bias scope.